Bare-Hand Humanoid Teleoperation for Hazardous Confined Environments:

The Remote Examination and Manipulation System (REMS)

System Concept and Evaluation Framework

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Empirical validation in progress. No human-subject results are reported.

Abstract

Hazardous confined environments—underground construction tunnels, subsurface utility networks, and nuclear decommissioning facilities—share a common and unresolved problem: the tasks that most urgently require human judgment are also the tasks that most dangerously expose human bodies. Inspection-capable quadruped robots can navigate these spaces and collect sensor data, but they cannot operate the human-scale controls— ventilation doors, isolation valves, circuit breakers, and pendant lever panels—that safety response demands. Fully autonomous systems cannot substitute for the certified professional judgment and legal attestation that regulations require.

We propose **REMS** (Remote Examination and Manipulation System): a single Apple Vision Pro headset streams bare-hand tracking and head pose to a Unitree G1 humanoid robot with Inspire Hand dexterous manipulators, with no handheld locomotion controller required. *Head-driven locomotion* keeps both hands continuously free for bimanual alarm response during route traversal—eliminating the mode-switch delay that joystick-based systems impose at the moment emergency intervention begins. REMS is validated in underground construction inspection as the primary scenario, encompassing atmospheric route examination, ventilation control verification, and post-alarm equipment isolation; the framework extends naturally to other hazardous confined environments where human-scale bimanual manipulation is required but human presence is dangerous. We present the system architecture and a proposed evaluation framework; controlled experiments are ongoing.

Keywords: hazardous confined environments, underground construction, humanoid robot, mixed reality, teleoperation, bare-hand tracking, head-driven locomotion, dexterous manipulation, zero-entry, REMS

1 Introduction

1.1 The Unresolved Problem: Expert Judgment in Dangerous Places

Across a wide range of safety-critical industries, the same asymmetry appears: the personnel most qualified to assess a hazardous environment are also the ones most exposed to it. In underground construction, a certified competent person must enter a newly blasted tunnel section before any crew may follow, confirming atmospheric safety, structural integrity, and equipment status [?]. In nuclear decommissioning facilities, radiation protection supervisors must physically access confined reactor spaces to verify isolation before maintenance crews proceed. In subsurface utility networks, qualified inspectors must enter confined spaces to confirm valve positions, atmospheric conditions, and structural soundness before any intervention.

The common thread is not the specific regulation but the underlying logic: professional attestation by a qualified person, based on direct observation and physical verification, is the legal and operational precondition for crew entry. Sensor networks and autonomous monitoring systems generate alarms; only a human expert generates the signed attestation that a workplace is safe.

This logic produces a structural exposure problem. The examination *requires* first entry, and first entry carries the highest risk of any activity in the cycle. Underground coal mining makes this asymmetry most explicit: before every shift, a *certified examiner*—known as a *fire boss*—must enter the mine alone, walk the haulage routes, test the atmosphere, confirm ventilation controls, assess roof stability, and inspect equipment, signing a legally binding Examination Record before any production worker may descend (30 CFR § 75.360) [1]. The fire boss is this paper's primary motivating case, but the problem they face is not unique to coal mining.

1.2 Why the Problem Persists

Three categories of existing technology address parts of this problem but none eliminates first-entry exposure.

Fixed monitoring systems. Atmospheric monitoring systems (AMS), SCADA sensor networks, and proximity detection systems (PDS) provide continuous readings at instrumented locations and warn when personnel approach operating equipment [2, ?]. These systems are valuable but structurally mismatched to the examination requirement. Fixed sensors cannot walk a route, sample atmosphere at locations where hazards accumulate but sensors are not installed, confirm that ventilation hardware is physically intact, or—after an alarm—execute post-alarm isolation.

Inspection-capable quadruped robots. Legged robots—most notably Boston Dynamics Spot—have been deployed in underground mines, construction sites, and industrial facilities for sensor-based route surveys, atmospheric monitoring, and visual inspection [3, 4, 5]. These platforms demonstrate that robots can traverse confined, GPS-denied environments and sustain shift-length inspection missions. However, they face a structural limitation that no sensor payload can resolve: they have no hands. The ventilation doors, valve panels, LOTO isolation points, circuit breakers, and pendant lever panels that safety examination requires to be *operated* were designed for human-scale bimanual kinematics. A quadruped can walk past these controls and log a reading; it cannot open a door to confirm airflow, turn a valve to isolate a line, or pull a lever to de-energize a machine.

Fully autonomous humanoid robots. Autonomous systems could in principle perform the physical tasks that quadrupeds cannot, but full autonomy cannot satisfy the regulatory requirement for certified human judgment and attestation. A licensed professional whose signature carries legal force cannot be replaced by an algorithm executing pre-programmed sequences.

The gap is therefore not "replace the expert with a better sensor" or "replace the expert with a smarter robot." It is: *preserve the expert's certified judgment and legal accountability while removing their body from first-entry exposure.*

1.3 Our Approach: REMS

We propose **REMS** (Remote Examination and Manipulation System): the expert stays on the surface, operates a humanoid robot through a mixed-reality interface, and signs the attestation record based on what they observed and directed—exactly as they would after a physical examination, but without first-entry exposure.

REMS is built around a single Apple Vision Pro headset with no additional handheld or body-worn hardware. The operator's bare-hand movements are tracked at 27 joints per hand by ARKit inside-out tracking and streamed to a Unitree G1 humanoid robot equipped with Inspire Hand dexterous manipulators. *Head-driven locomotion* maps the operator's head pitch and yaw to robot walking and turning commands, leaving both hands continuously free.

The hands-free locomotion design is not a convenience feature. In hazardous confined environments, alarms arrive without warning and require immediate bimanual response—without releasing a locomotion joystick, without a mode switch, without the cognitive cost of transitioning between control modalities. Head-driven locomotion ensures that both hands are always ready, whether the robot is stationary or mid-transit when an alarm occurs.

We validate REMS in underground construction inspection as the primary scenario—a domain where competent person requirements, atmospheric hazards, and human-scale equipment controls all converge. The system architecture and evaluation framework are designed to generalize to other hazardous confined environments with analogous task structures.

1.4 Contributions

1. A problem framing showing why first-entry exposure in hazardous confined environments cannot be eliminated by sensors, quadruped robots, or autonomous systems alone, and why human-in-the-loop teleoperation with a humanoid platform is the minimal configuration that closes the gap (§1–3).
2. **REMS**: a single Apple Vision Pro operator stack with no additional hardware that streams bare-hand tracking and head pose to a full humanoid robot with dexterous hands, enabling the complete examination and manipulation task scope that hazardous confined environments demand (§4).
3. A *head-driven locomotion* interface that decouples robot walking from hand control, keeping both hands free for bimanual alarm response throughout route traversal, with joystick teleoperation as an experimental baseline (§4).
4. A proposed evaluation framework spanning three task classes—atmospheric route examination, ventilation and access control, and post-alarm equipment intervention—validated in underground construction inspection and designed to generalize to analogous hazardous environments (§5).

We do **not** claim to eliminate the legal role of certified examiners, achieve intrinsically safe deployment in explosive atmospheres, or demonstrate experimental superiority of head-driven locomotion for static tasks. We **do** claim that REMS lets the expert perform their complete statutory duty from the surface—preserving the professional judgment and attestation that the law requires, while eliminating first-entry exposure. Controlled experiments are ongoing; this preprint establishes the system concept and evaluation framework.

2 Background

2.1 Hazardous Confined Environments: Common Task Structure

Despite differing regulatory frameworks and physical characteristics, the hazardous confined environments that motivate REMS share a common task structure. A qualified person must physically enter, walk a route or access confined spaces, assess atmospheric and structural conditions, verify the operational status of safety-critical equipment through physical interaction, and produce a signed record attesting that the environment is safe for subsequent crew entry.

Underground construction (tunneling). OSHA 29 CFR § 1926.800 requires that a competent person examine the underground work area before each shift, testing for hazardous gas concentrations, confirming ventilation operation, and assessing ground stability [?]. Haulage entries are typically 3–6 m wide and 2–3 m high, with uneven floors, cable runs, and equipment creating irregular navigation challenges. Following blasting operations, atmospheric conditions can change rapidly and substantially; the competent person must enter before production crews regardless of conditions at the previous examination.

Underground coal mining. 30 CFR § 75.360 requires a certified examiner (fire boss) to conduct a preshift examination encompassing atmospheric sampling, ventilation verification, roof and ground assessment, and conveyor inspection before every shift [1]. The fire boss's signed Examination Record is the legal precondition for shift start. Table 1 maps the six interdependent task classes to their governing regulations; this scenario provides the primary regulatory reference for REMS because its examination scope is most fully codified.

Nuclear decommissioning. Radiation protection supervisors must access confined reactor spaces, valve galleries, and containment areas to verify isolation before maintenance crews enter. These environments combine confined geometry with radiation fields that impose strict time limits on human exposure, creating acute pressure to minimize personnel presence while maintaining expert oversight.

Common manipulation requirements. Across all three scenarios, the tasks that most require physical interaction involve the same category of hardware: ventilation doors, valve panels, isolation switches, emergency stops, and pendant lever panels. These controls were designed for human-scale bimanual kinematics and are positioned at human-scale heights. A system that can operate these controls in one scenario can, with appropriate configuration, operate analogous controls in the others.

2.2 Preshift Examination: Statutory Scope (Underground Coal Reference)

Table 1 maps the six task classes of the underground coal preshift examination to governing regulations and required physical capabilities. This table serves as the primary reference for REMS task design because the coal mining examination is the most fully codified instance of the general hazardous confined environment examination pattern. The ventilation verification, post-alarm intervention, and route traversal requirements have direct analogues in underground construction and nuclear decommissioning.

A critical operational feature of actual examination practice is that these task classes are not performed sequentially. The examiner walks the route *once*, performing atmospheric sensing, visual assessment, and general hazard observation simultaneously, pausing only at specific locations for point actions. Any teleoperation system that replicates this function must support concurrent multi-task operation during traversal.

2.3 Physical Environment Constraints

Hazardous confined environments impose constraints on robotic deployment that surface-optimized systems do not face.

Confined geometry. Haulage entries, tunnels, and plant corridors are sized for human-operated equipment rather than large robotic platforms, with irregular obstacles requiring agile navigation. Safety-critical controls are positioned at human-scale heights and designed for human-scale hand geometry; a robot must match these envelopes to operate them without facility modification.

GPS-denied communication. Robot localization must rely on onboard sensing (LiDAR, IMU) or pre-mapped environments. Radio frequency propagation is severely attenuated; high-frequency signals require dense repeater infrastructure. Private 4.9G/5G underground networks are an established solution [6, 7]; we treat communication infrastructure as a deployment prerequisite rather than a research contribution.

Atmospheric and ignition hazards. Electrical equipment in gassy environments must meet intrinsic safety standards. Consumer electronics, including the Apple Vision Pro, are not certified for use in potentially explosive atmospheres. REMS places the Vision Pro at the surface operator station only; only certifiable equipment accompanies the robot underground.

2.4 Existing Technology: Capabilities and Gaps

Table 2 summarizes the capability landscape. Fixed monitoring systems address continuous sensing but cannot perform route-based examination or post-alarm physical intervention. Quadruped inspection robots can

Table 1: Preshift examination duties (underground coal reference), governing regulations, and required physical capabilities. Analogous task classes exist in underground construction (OSHA 29 CFR § 1926.800) and nuclear decommissioning.

Task class	Regulation	Required physical capability
Atmospheric sampling	§§ 75.360, 75.323	Locomotion along route; sensor positioning at prescribed heights; judgment on reading interpretation
Ventilation verification	§ 75.326	Physical inspection of stoppings and overcasts; door opening and closing; airflow measurement
Roof and ground assessment	§§ 75.202, 75.220	Continuous visual inspection; close inspection at flagged locations; identification of unstable material
Conveyor and electrical	§§ 75.1100, 75.1731	Visual inspection of belt components; fire suppression device confirmation
Post-alarm intervention	§§ 75.323, 75.1501	Bimanual LOTO at lever panels; equipment isolation; crew-entry blocking; emergency retreat
Examination Record sign-off	§ 75.360(f)(g)	Professional judgment synthesis; certified written attestation

Table 2: Capability comparison.

	AMS/ PDS	Quad- ruped	Pendant RC	REMS
Continuous gas sensing	✓	✓	—	via robot
Route-based examination	—	partial	—	✓
Ventilation verification	—	visual	—	✓
Roof assessment	—	visual	—	✓
Bimanual manipulation	—	—	✓	✓
Post-alarm LOTO	—	—	partial	✓
Standoff lever control	—	—	—	✓
Certified sign-off	—	—	✓	✓*
Zero first-entry exposure	—	✓	—	✓
Hands-free in transit	—	N/A	—	✓

*Surface operator retains full legal accountability.

traverse confined environments and collect sensor data but structurally cannot operate human-scale controls. Pendant remote controls enable equipment operation but place the operator's body within hazardous proximity of operating machinery. REMS closes the gap by combining human expert judgment, remote presence via MR, and a humanoid robot with dexterous hands—the minimal configuration that satisfies all task classes while preserving regulatory accountability.

3 Related Work

Prior work addresses individual components of the REMS problem space—mixed-reality humanoid control, inspection robot deployment, dexterous hand retargeting, and sparse-sensor locomotion inference—but no existing system integrates these capabilities into a framework targeting the certified-examiner function in hazardous confined environments.

3.1 Mixed-Reality Humanoid Teleoperation in Construction

Zhang and Jebelli have published an extensive series on MR humanoid teleoperation for above-ground construction tasks, including context-aware spatial overlays, assistive autonomy frameworks, and user studies on raised-floor installation [8, 9]. This work establishes that MR-guided humanoid teleoperation can support contact-rich manipulation in structured environments. Three structural differences separate this body of work from REMS. First, existing systems target above-ground productivity workflows, not life-safety examination in GPS-denied, atmospherically hazardous environments. Second, operator interfaces typically combine the MR headset with additional input devices; REMS uses the Vision Pro as the sole interface with no handheld locomotion controller. Third, alarm-interleaved locomotion-manipulation scenarios have not been the focus of prior evaluation.

3.2 Quadruped and Legged Robots in Confined Environments

Quadruped robots have seen growing deployment in underground environments for inspection and monitoring. Boston Dynamics Spot has been fielded at Glencore Kidd Creek [4], LKAB Kiruna [3], and multiple Australian mining operations [5], demonstrating shift-length endurance in GPS-denied environments. Autonomous navigation stacks for underground legged robots have also been demonstrated in fully unlit conditions [10].

These deployments establish the operational viability of legged robots in confined environments but stop short of the manipulation tasks that safety examination requires. The structural absence of hands means that ventilation door operation, valve isolation, and LOTO execution remain outside their capability envelope. REMS accepts the relative bipedal stability cost in exchange for the dual-hand reach envelopes that the full examination scope demands.

3.3 Dexterous Humanoid Teleoperation Interfaces

Recent work has established that Apple Vision Pro can serve as a high-fidelity teleoperation interface for humanoid robots. VisionProTeleop demonstrates real-time ARKit-based hand and head pose streaming to robot arms [11]. Open-TeleVision adds active stereo visual feedback for immersive upper-body retargeting [12]. Bunny-VisionPro targets bimanual dexterous manipulation for imitation learning dataset collection [13]. SONIC provides a scalable whole-body motion tracking controller trained on large-scale human motion capture, unifying locomotion and upper-body control through a 3-point teleoperation interface [14].

These systems establish the technical feasibility of bare-hand humanoid teleoperation but have been evaluated in laboratory manipulation tasks, not route-traversal scenarios with alarm-driven bimanual intervention requirements. Critically, all existing implementations either omit legged locomotion or delegate walking to a separate joystick or foot pedal, creating the locomotion-to-manipulation mode switch that REMS is designed to eliminate.

3.4 Full-Body Pose Estimation from Sparse Headset Inputs

AvatarPoser reconstructs whole-body pose from head and hand tracking using a Transformer encoder with IK refinement at over 600 fps [15]. EgoPoser extends this to intermittent hand observations and diverse body morphologies [16]. Both systems target virtual avatar embodiment: they infer how a user's lower body *likely moved* for visual presence.

REMS uses head pose differently: as a *deliberate control signal* that commands robot locomotion. Head pitch and yaw become walking and turning references, closed-looped with the robot's onboard IMU. Neither approach requires pelvis trackers or handheld locomotion controllers beyond the headset's native tracking, but they diverge in intent: AvatarPoser asks *"what was the user doing?"* while REMS asks *"what should the robot do?"*

Summary. No prior work defines a teleoperation framework that addresses the complete examination duty scope of a certified expert in a hazardous confined environment—spanning atmospheric sensing, ventilation verification, bimanual equipment intervention, and attestation record sign-off—on a single MR operator interface with hands-free locomotion designed for alarm-interleaved scenarios. Table 2 summarizes the resulting capability gap that REMS targets.

4 System Architecture

We instantiate the REMS framework on a single teleoperation stack that couples a surface-side mixed-reality operator station to a human-scale bipedal robot, deliberately reserving both of the operator's hands for manipulation by driving locomotion from head motion instead of a handheld controller. Figure 1 shows the end-to-end signal flow. A reference implementation is released as open source [17]; here we describe the architecture at the level needed to reproduce the design rather than the code.

4.1 Hardware Stack

Operator station. The operator remains at the surface wearing an Apple Vision Pro. Inside-out tracking supplies a continuous 6-DoF head pose together with articulated hand skeletons of up to 27 joints per hand, from which REMS consumes the head transform, both wrist poses, and per-finger joint chains [11, 12]. No handheld controller, glove, exoskeleton, or teleoperation booth is required. The headset is not certified for use in potentially explosive atmospheres and never enters the hazardous environment; it is a surface console only.

Robot platform. The field agent is a Unitree G1 bipedal humanoid in a 29-DoF whole-body configuration, fitted with bilateral Inspire dexterous hands. Each hand exposes six actuated channels— flexion of the little, ring, middle, and index fingers plus thumb bend and thumb rotation— which is the resolution constraint that shapes retargeting (§4.2). Proprioception comprises joint encoders and a pelvis-mounted IMU whose orientation is fed back to the operator side.

Sensing payload. Multi-gas sensing (CH_4, O_2, CO) and LiDAR mapping enter the architecture as payload interfaces rather than fixed components, so that the same control logic can be driven by physical sensors or, as in current validation, by scripted virtual sensor fields.

Communication. Operator station and robot share a local IP network carrying command, feedback, and video return channels. Sustaining these over kilometers of underground infrastructure is an independent deployment problem addressed by leaky feeder or private cellular installations [6, 7], orthogonal to the control mapping described below.

4.2 Software Pipeline

Control loop. A teleoperation bridge converts streamed operator poses into robot-frame references and publishes whole-body targets at 50 Hz ($T = 20\,\mathrm{ms}$). Each frame carries a locomotion command (mode, travel direction, facing direction, speed, pelvis height) together with upper-body targets for the head and both wrists.

Coordinate frames. Headset poses arrive in a Y-up convention; the humanoid uses Z-up. Writing T_{hmd} for a raw 4×4 operator pose, the robot-frame pose is

$$T_{\mathrm{rob}} = T_{R\leftarrow X}\, T_{Z\leftarrow Y}\, T_{\mathrm{hmd}}\, T_{X\leftarrow R}, \tag{1}$$

where $T_{Z\leftarrow Y}$ resolves the up-axis convention and $T_{R\leftarrow X}$ maps the tracking frame onto robot axes. Wrist targets are additionally composed with a per-arm end-effector

Table 3: REMS dual-protocol framework: task class, substitution target, and primary benefit.

Pillar	Task class	Replaces	Primary benefit
I: RE-TEP	Remote preshift examination	Certified examiner first entry	Zero human entry into unknown atmosphere; tiered response
II: REMP	Standoff lever manipulation	In-zone pendant operator	Zero human pinch-zone exposure; sacrificial robot

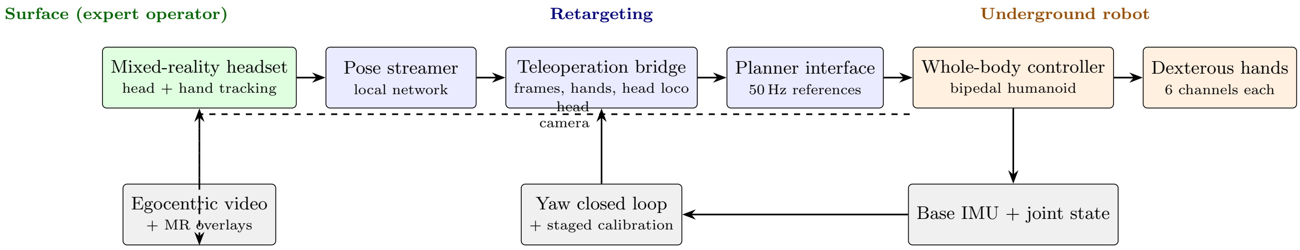

Figure 1: REMS architecture. Head and hand poses captured at the surface are retargeted into whole-body references issued at 50 Hz to the robot's learned controller. Dashed arrows denote the egocentric video return path; the lower solid path carries proprioceptive feedback for calibration and closed-loop yaw stabilization.

convention matrix so that a supinated operator palm produces a supinated robot palm.

Whole-body control. Locomotion and upper-body tracking are executed by a learned whole-body controller of the SONIC family [14]. REMS acts as a retargeting and protocol layer; the robot remains responsible for footstep placement and balance on uneven floor.

Hand retargeting. The 27 tracked joints per hand are reduced to a six-dimensional command

$$\mathbf{u} = [u_{\text{lit}},\ u_{\text{rin}},\ u_{\text{mid}},\ u_{\text{ind}},\ u_{\text{t,bend}},\ u_{\text{t,rot}}]^{\top} \in [0,1]^6. \tag{2}$$

For each non-thumb finger, metacarpophalangeal, proximal, and distal interphalangeal angles are combined as a weighted sum, yielding a robust flexion scalar even under partial occlusion. Each channel is normalized against a per-operator calibration, low-pass filtered ($\alpha = 0.45$), and mapped linearly onto the hand's command range.

Latency. The command loop runs at 50 Hz on a local network. A controlled end-to-end latency measurement from operator motion to robot motion and back to video return is part of the evaluation program in §5.

4.3 Head-Driven Locomotion

Mapping. After calibration, the operator's horizontal head displacement is differentiated to give a forward and lateral velocity pair; head yaw sets the facing direction independently:

$$\mathbf{m} = \frac{\dot{\mathbf{p}}_{xy}}{\|\dot{\mathbf{p}}_{xy}\|}, \qquad \mathbf{f} = [\cos\psi,\ \sin\psi,\ 0]^{\top}, \tag{3}$$

with travel direction $\mathbf{m}$ from head velocity and facing $\mathbf{f}$ from smoothed relative head yaw ψ. Motion is gated by a deadzone of 0.07 m/s, limited to 0.45 m/s and 0.35 rad/s, with first-order smoothing ($\alpha \approx 0.12$). A vertical head drop optionally commands a reduced pelvis height for low-clearance inspection.

Yaw stabilization. Commanded facing is closed against measured base orientation to prevent slow heading drift in corridor-following tasks.

Calibration. Engaging control is staged: the operator holds a forearms-forward reference pose for two seconds; the whole-body policy engages; arm targets are matched to the robot's measured configuration; live teleoperation begins. Head facing and squat height can be re-zeroed mid-route without discarding arm calibration.

Relevance to hazardous examination. In hazardous confined environments, alarms arrive while the robot is still in transit. A handheld controller occupies one hand and imposes a mode switch before bimanual response can begin. Head-driven locomotion removes walking from the manual channel entirely, so both hands are available throughout transit. For fully static manipulation, where the robot has stopped before the hands engage, no advantage over a handheld baseline is hypothesized; we state this as a testable null result in §5.

4.4 Mixed-Reality Operator Interface

The operator's primary view is an egocentric video feed from the robot's head camera (1280×720, 30 fps). REMS composites protocol state onto this feed: atmospheric tier indicators, standoff boundary markers, proximity cues, and control-panel labels. Examination artifacts are timestamped as the route proceeds so that the attestation record is compiled from the session rather than reconstructed afterward.

4.5 Simulation Environment

Development and rehearsal use a MuJoCo model of the humanoid with both dexterous hands, exercised through the same bridge and calibration procedure as hardware. The environment includes an instrumented haulage route for examination tasks and a pendant-lever-style control panel with an explicit standoff line for isolation tasks.

Three sim-to-real gaps bear on our claims. Contact friction at real levers is stiffer than simulated equivalents, so simulation success should be read as an upper bound. Bipedal locomotion on wet, uneven floors remains less proven than quadruped locomotion in deployed underground systems. No laboratory trial can reproduce explosive atmospheres; physical validation uses de-energized mock-ups and virtual gas thresholds.

5 Evaluation Framework

REMS makes two claims requiring empirical validation: that a single-MR-headset humanoid teleoperation system can execute the full examination task scope with sufficient reliability and speed to be operationally meaningful; and that head-driven locomotion reduces alarm-response latency in interleaved locomotion–manipulation scenarios compared to joystick-based control.

We evaluate these claims through three task classes validated in underground construction inspection as the primary scenario. The task structure is designed to generalize to analogous hazardous confined environments.

5.1 Task A: Atmospheric Route Examination

Scenario. A simulated underground construction entry contains five atmospheric sampling waypoints, including one elevated location requiring sensor arm elevation to the simulated roof line. The operator navigates using head-driven locomotion and completes a designated sample at each waypoint.

Metrics. Waypoint completion rate (%); sensor positioning accuracy (cm, target ± 5 cm); route completion time (min); NASA-TLX.

Expected result. Baseline traversal and sensing competence. No locomotion-interface advantage hypothesized; this task class is also within the envelope of quadruped platforms and serves as a competence baseline.

5.2 Task B: Ventilation Door Operation

Scenario. A simulated ventilation door at human-scale height (1.0–1.2 m) requires a pull-open, hold, and push-close sequence. A second condition requires the operator to open the door, pass the robot through, and close it—a sequential bimanual task with no analogue in quadruped capability.

Metrics. Task success rate (%); completion time (s); grip stability (% of grasps maintained through full sequence); NASA-TLX.

Expected result. Primary evidence that REMS closes the manipulation gap that quadruped platforms leave open.

5.3 Task C: Post-Alarm Equipment Intervention

C1 — Static condition. Robot positioned at standoff distance from a simulated control panel. Operator executes a three-element isolation sequence (lever, switch, confirmation) within a 30-second window. No locomotion-interface difference hypothesized (H_0, testable null result).

C2 — Interleaved condition. Operator navigates the robot along a corridor; at a randomized point, an alarm fires. Operator must halt, navigate to the control panel, and complete the isolation sequence.

In the joystick condition, releasing the controller before reaching the panel is required. In the head-driven condition, both hands are already free.

Metrics. Alarm-to-first-contact time (s, primary outcome); isolation sequence completion time (s); sequence error rate (%); NASA-TLX.

Primary hypothesis. Head-driven locomotion produces significantly shorter alarm-to-first-contact times than joystick control in C2 (H_1).

5.4 Experimental Design

Participants. $N = 16$ participants recruited from university engineering departments, with no prior experience operating humanoid robots. Sample size is appropriate for a proof-of-concept study; we do not claim statistical power for small effects.

Design. Tasks A and B: within-subjects across both locomotion conditions, counterbalanced. Task C: C1 within-subjects; C2 between-subjects to prevent strategy transfer.

Environment. All trials in MuJoCo with the G1 + Inspire Hand model, in a laboratory corridor approximating underground construction entry geometry (ceiling 2.2 m, width 3.0 m). Laboratory performance does not directly transfer to operational underground deployment; the evaluation establishes proof-of-concept capability and quantifies the locomotion interface effect under controlled conditions.

6 Discussion

6.1 The Expert Remains Central

REMS does not replace the certified examiner. It relocates them. The operator retains every element of the regulatory role: they walk the route (through the robot), apply professional judgment to observations, make escalation decisions, execute required interventions, and sign the attestation record. What changes is only the physical location of their body.

REMS is a telepresence system that extends the expert's physical reach while preserving their cognitive and legal centrality. The attestation record signed after a REMS examination is signed by the same certified person who would have signed it after a physical examination; the legal chain of custody is unbroken.

6.2 Bipedal Stability vs. Quadruped Platforms

The choice of a bipedal humanoid accepts a real engineering cost: quadruped robots demonstrate superior stability on irregular terrain and have been operationally validated in underground environments over extended missions. We do not contest this advantage.

The relevant trade-off is task coverage. Quadruped platforms can perform Task A (atmospheric route examination); they cannot perform Task B (ventilation door operation) or the isolation component of Task C without hands. Bipedal platforms accept instability risk in exchange for the dual-hand reach envelopes the full examination scope demands.

6.3 Limitations

Intrinsic safety certification. Consumer electronics, including the Unitree G1 and Inspire Hand, are not certified for use in potentially explosive atmospheres. REMS as currently designed is appropriate for evaluation in simulation and non-gassy environments; operational deployment in classified hazardous areas requires a separate certification program.

Force feedback absent. Ventilation door and valve operations benefit from tactile confirmation of contact and applied force. The current system provides visual feedback only; a force-feedback pipeline is identified as a priority for future work.

Physical scaling not supported. REMS can identify and flag unstable material; physical dislodgement requires impact force and tool manipulation with force feedback beyond current system capability.

Communication infrastructure. Low-latency wireless connectivity is assumed as a deployment prerequisite. Communication loss mid-examination requires a fallback protocol not yet formalized in REMS.

Regulatory acceptance pathway. Whether REMS-conducted examinations satisfy regulatory attestation requirements in each target domain requires engagement with the relevant authorities before operational deployment. The system satisfies the functional requirements in principle; administrative acceptance is a separate process.

6.4 Future Work

Immediate next steps: controlled experiments with the framework in §5; end-to-end latency measurement under realistic network conditions; Inspire Hand grasp reliability on representative hardware (door handles, valve panels, lever panels).

Longer-term: force-feedback pipeline for contact-rich manipulation; semi-autonomous waypoint navigation to reduce operator workload during routine traversal; integration of onboard sensor arrays with automatic tier-classification display; regulatory engagement in each target domain.

7 Conclusion

Hazardous confined environments demand that the most qualified people enter the most dangerous places. Fixed sensors cannot walk a route or operate equipment. Quadruped robots can walk a route but cannot operate equipment. Autonomous systems can do both but cannot provide the certified human judgment and attestation that regulations require.

REMS closes this gap by keeping the expert in the loop—at the surface, behind a mixed-reality interface, directing a humanoid robot that performs the physical examination on their behalf. The expert walks the route, reads the atmosphere, verifies the ventilation, operates the controls, and signs the record—as they always have, but without exposure to the hazards that make the examination necessary.

The key enabling design is the decoupling of locomotion from manipulation. Head-driven locomotion frees both hands throughout route traversal, so that when an alarm fires mid-route, the operator's hands are already ready for the bimanual response that the alarm demands. This is not a convenience—it is a safety-critical design requirement in environments where alarm response time has regulatory meaning.

REMS is a concept and system architecture preprint; controlled experiments are ongoing. We offer it as a framework for discussion, evaluation, and eventual regulatory engagement that could let certified experts do their job—completely, compliantly, and safely— without ever entering the hazard zone.

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[17] Authors TODO. Vision pro G1 inspire teleoperation stack. Open-source implementation, https://github.com/TODO/visionpro-g1-inspire-teleop, 2026. Replace URL before arXiv submission.